

Andy Chiem

(267) 690-7657 | ac3436@cornell.edu | [in/andychiem2006](https://in.andychiem2006) | **Secret Clearance (Active)**

EDUCATION

Cornell University

Bachelor of Science in Electrical and Computer Engineering | GPA: 3.7/4.0

Ithaca, NY

Expected Graduation: May 2029

EXPERIENCE

Woodward McCoach Inc.

FPGA Development Engineer Co-op

West Chester, PA

Mar 2026 – Aug 2026

- Developed open-source 100G Ethernet verification suite in Python and cocotb as sole contributor, from packet generation through 64b/66b encoding, virtual lane distribution, and marker insertion to Versal DCMAC interface
- Assembled GTM-to-GTYP SerDes simulation in SystemVerilog, resolving a bit-order incompatibility in proprietary chip-to-chip protocol and unblocking high-speed serial communication across transceiver families on production hardware
- Resolved board-specific discrepancies in AXI-Lite bus utilization analysis using Vivado ILAs, increasing measured bandwidth from 70% to 100% of expected under 100GAUI stimulus on production hardware
- Engineered IEEE 802.3-compliant virtual MAC layer in Scapy implementing preamble insertion, interframe gap, and frame check sequence calculation, enabling end-to-end Ethernet stimulus across the verification stack

Drexel VLSI and Algorithms Lab

Research Assistant | *Unconventional Computing*

Philadelphia, PA

Nov 2025 – May 2026

- Designed an oscillator-based Ising-model optimizer for FPGA in SystemVerilog, implementing annealing and direct calculation methods to solve combinatorial optimization problems with 94% accuracy across 32 nodes
- Evaluated 20+ behavioral simulations in Questa/ModelSim to determine optimal piecewise approximation of sinh/cosh coupling functions, balancing solution accuracy against hardware resource utilization

Pennsylvania Army National Guard

Cadet, Electromagnetic Warfare Track

Horsham, PA

Mar 2024 – Present

- Train in electromagnetic warfare operations and signal intelligence systems through inactive duty training cycles

PROJECTS

FPGA Options Greeks Accelerator | [GitHub](#)

June 2026 – Present

- Build an end-to-end FPGA accelerator for Zybo Z7 SoC that ingests live market data over UDP/Ethernet and computes Black-Scholes option Greeks in hardware, achieving accuracy within 1E-4 of reference calculations
- Reduce hardware resource cost for transcendental functions by replacing floating-point units with a parameterized CORDIC core supporting circular, hyperbolic, and linear modes in Q16.16 fixed-point
- Accelerate option delta computation to ~63-cycle deterministic latency (630ns at 100MHz) by implementing a multi-stage SystemVerilog compute chain and pipelining submodules

Retro Display Controller | [GitHub](#)

June 2025

- Implemented a SystemVerilog video pipeline that drove HDMI output at 60 Hz, generating VGA signals, encoding with transition-minimized signaling, and rendering retro-style text on external monitor
- Designed a synthesizable FSM in SystemVerilog modeling vending machine control logic as a display backend, featuring debounced real-world inputs and item selection state transitions
- Executed synthesis and implementation using Vivado for Zybo Z7 SoC, validating RTL across 20+ simulations

Single-cycle RISC-V CPU

June 2025

- Built a single-cycle RISC-V architecture CPU in SystemVerilog, implementing a subset of the RV32I instruction set
- Validated full-system behavior using Verilator and cocotb co-simulation, achieving zero functional mismatches

SKILLS

Technical: RTL design, timing analysis, embedded systems, hardware validation, signal interfacing, PCB design

Tools: Vivado, Questa/ModelSim, Altium Designer, cocotb, Git, Linux, STM32CubeIDE

Languages & Protocols: SystemVerilog, Verilog-2001, C, C++, Python, MATLAB, bash, Tcl, Ethernet (1G/100G), AXI-Stream, AXI-Lite, UDP/IP, I2C, UART